Q1 Study Guide

Feeding America's need estimates vs. The Food Group's food shelf activity

MinneMUDAC 2026 · The Food Group challenge · data years 2022–2024

What this is. Everything we found for Question 1, explained in plain words, with the numbers that back each claim and a short note on how we got them. Use it to prepare the presentation and to answer judges' questions.

How to read it. Section 1 is the 60-second answer. Sections 2–3 cover the data and the key terms. Sections 4–7 go through the four parts of Q1. Sections 8–10 cover limits and recommendations. Section 11 is a bank of likely questions with answers, and section 12 is a suggested slide outline.

Where the full work lives. notebooks/q1_need_vs_foodshelf_activity.ipynb (section numbers in brackets, e.g. [§31], point to it). Saved outputs: Data/processed/q1_county_profiles.csv, q1_county_mismatch_summary.csv, q1_county_year_mismatch.csv.

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Appendix: numbers at a glance

1. The 60-second answer

Question 1 asks four things. Here is the short answer to each.

Sub-question	Answer	Key evidence
1. Do need estimates and food shelf activity tell the same story?	**Partly.** They agree on county *size* and on the *direction* over time, but not on *intensity*.	Totals: ρ ≈ 0.9 (but plain population does just as well). Per resident: ρ ≈ 0 by 2024. 2022→2024: need +25%, visits +46%, individuals +60%.
2. Which metric correlates most / least?	**Pounds** is consistently the strongest; **Individuals** is usually the weakest (Visits nearly tied). The differences are small.	Pounds is highest in every year and both correlation types. Bootstrap intervals overlap, so the lead is consistent but not large.
3. Where does the relationship break down?	12 counties are persistently ≥2× above or ≤½ of what similar-need counties record.	**Lower:** Nicollet, Pope, Lac qui Parle, Norman, Roseau. **Higher:** Steele, Washington, Rice, Renville, Mahnomen, Pipestone, Olmsted.
4. What county characteristics might explain it?	**Geography, access and the food bank network**, more than hardship.	Strongest factor: % of residents whose nearest food shelf is in another county (ρ = −0.33, p = 0.002). Food bank network p = 0.02. Suggestive: no "lower" county has a mobile pantry. Poverty, vehicles, rural share, SNAP: no association.

One-sentence takeaway: County need estimates alone are a weak guide to where food shelf activity happens. **Where food shelves are, how big they are, and how far people must travel** shape the recorded numbers at least as much as estimated need, and because activity is counted where the food shelf sits, a county can look under-served when its residents are being served next door.

2. The data we used

Data	What it is	Used for
Feeding America, Map the Meal Gap (MMG 2025 and 2026 releases)	A **model estimate** of food-insecure people (*fi_persons*) and the food insecurity rate (*fi_rate*) per county per year. Not a head count.	The "need" side. Years 2022, 2023, 2024.
The Food Group food shelf reports (org_FSStats, cleaned)	Monthly reports from each food shelf site: households (Visits), pounds, individuals (adults + children + seniors).	The "activity" side. Summed to county × year. Meal programs excluded.
Census ACS 5-year	Population, poverty rate, households with no vehicle (with margin of error).	Per-resident rates; Part 4 characteristics.
Census 2020	Urban/rural population; population-weighted centers of 4,706 block groups.	Rural share; distance to the nearest food shelf.
Organization.xlsx + Census TIGERweb	Each site's Census block → coordinates; SuperShelf status.	Where food shelves are.
MN SNAP by county agency	Monthly SNAP cases/people by county agency.	SNAP participation (74 counties usable).
Census commuting flows 2016–2020	Workers by home county → work county.	Commuting test.
Helpline call logs (Incoming Call.xlsx, cleaned)	Calls with county, reason and referrals.	Supporting evidence that people are referred across county lines (Wilkin).

Why 2022–2024 only?

MMG has county estimates for 2022, 2023 and 2024. The food shelf data runs Jan 2022 – Jul 2026. The overlap is 2022–2024, and all three years have 12 full months of food shelf data. (We also never use August 2026 data, which the brief forbids.)

Data quality steps that matter for Q1

- **Typos removed, not "fixed".** E.g. Waseca, July 2023 reported 273,385 households; set to missing. Large-but-plausible values were kept and flagged. All 430 actions are in the issues log.
- **Blank is not zero.** A blank month means nobody reported, not that nobody came.
- **Incomplete county-years are flagged** (7 of them, e.g. Grant 2022 with 61% of site-months missing) and we checked that removing them changes no conclusion (largest change in any correlation: 0.027).
- **Wilkin County has no food shelf in the data**, so it can't be scored. It is kept visible, not silently dropped.
- **Joins use FIPS codes, never county names**, and every join was checked: 261 of 261 county-years matched.
- **Population check:** MMG's implied population (fi_persons ÷ fi_rate) is within ~2% of the Census population, so the Census population is a fair denominator.

Important caveat about "Individuals": it is **not unique people**. Someone who visits every month counts 12 times a year. So yearly "individuals served" can exceed the number of food-insecure people without being a contradiction.

3. Key terms in plain words

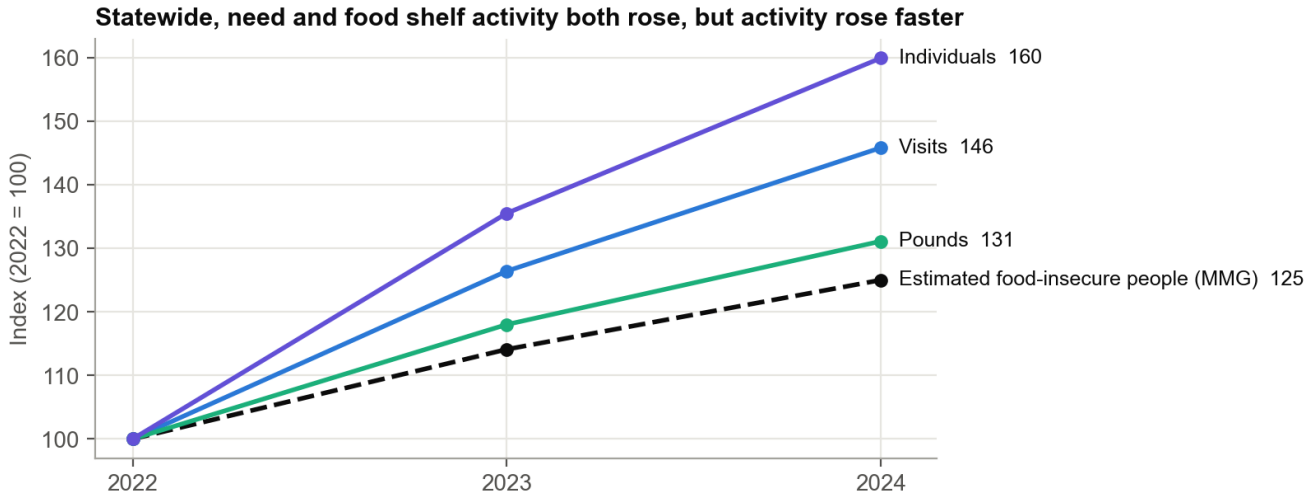
Term	Plain meaning	How to say it out loud
Correlation	How closely two things move together, from −1 to +1. 0 = no relationship.	"Counties high on one tend to be high on the other."
Spearman ρ (rho)	Correlation of **ranks**. One giant county (Hennepin) can't dominate it.	"Do counties keep the same order on both lists?"
Pearson r	Correlation of actual values (straight-line fit). Sensitive to extreme counties.	"Do points fall on a straight line?"
p-value	How surprising the result would be if there were really no relationship. Below 0.05 = unlikely to be chance.	"There's about a 2% chance of seeing this by luck."
Per resident	Activity ÷ population × 1,000. Removes county size.	"Visits per 1,000 residents."
Expected activity	What counties with a similar number of food-insecure people typically record (a fitted line, see §6).	"What a county like this usually records."
Gap / mismatch	Actual ÷ expected, on a doubling scale: 2× = one step up, ½ = one step down.	"Steele records 4 times what similar counties record."
Persistent	Flagged (≥2× or ≤½) in the same direction in at least 2 of 3 years, never the opposite.	"Not a one-year blip."
Bootstrap	Re-draw the 86 counties at random 2,000 times and recompute, to see how stable a result is.	"We re-ran it 2,000 times on reshuffled data."
Leave-one-out	Drop each county in turn and recompute, to find counties that drive the result.	"No single county is carrying this." (or: "Mahnomen is.")

Term	Plain meaning	How to say it out loud
Regression, R^2	Tests several factors at once, holding the others constant. R^2 = share of the variation explained (0.22 = 22%).	"Together, these explain about a fifth."
Mann-Whitney / Kruskal-Wallis	Rank-based tests: do 2 groups (or 3+ groups) differ?	"Is the lower group really different from the rest?"

4. Part 1: Do the two sets of numbers tell the same story?

Answer: partly. Same story on size and direction, different story on intensity.

Finding 1: Over time, both rose, but activity rose faster



All counties summed per year, indexed to 2022 = 100. Source: notebook [§6.1]. MMG county numbers don't sum exactly to the state total, so this is approximate.

- Need +25%, visits +46%, pounds +31%, individuals +60% (2022 → 2024).
- **Same direction, different size.** Individuals grows fastest partly because it counts repeat visits.

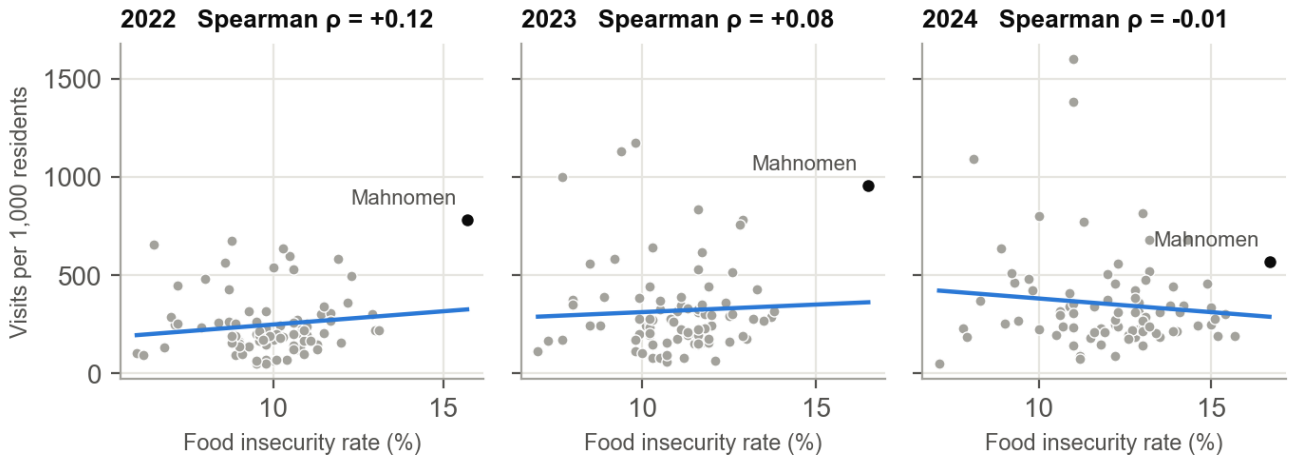
Finding 2: By county size, they agree strongly, but that's mostly population

Totals correlate strongly (Spearman ≈ 0.89–0.92). **But:** food-insecure people and plain population rank counties almost identically (ρ ≈ 0.99), and **population alone predicts activity just as well as MMG does** (e.g. 2024 visits: 0.90 with MMG vs. 0.91 with population). So "big counties have more of everything" explains most of this agreement.

Finding 3: By intensity (per resident), they don't agree

Counties with a higher food insecurity *rate* do not have more food shelf activity *per resident*. Spearman per resident: 0.12 / 0.08 / −0.01 for visits (2022 / 2023 / 2024), fading to about zero.

Per resident, a higher need rate does not mean more food shelf visits



Each dot is a county (86 per year). Blue line = simple straight-line trend; it is almost flat and tilts downward by 2024. Pounds and individuals look the same (per-resident Spearman 0.20 / 0.09 / 0.02 and 0.10 / 0.06 / 0.00). Mahnomen, the county with the highest rate, is the main point pulling the early years up. Source: q1_county_year_mismatch.csv [§6.3, §11].

How to read it: if need and activity told the same story per resident, the dots would climb from lower-left to upper-right. Instead they form a flat cloud: counties at 8% and at 14% food insecurity have roughly the same visits per 1,000 residents. None of the per-resident correlations is statistically significant except Pounds 2022 (Pearson), and that one depends on Mahnomen alone.

Finding 4: The three metrics agree with each other

Per resident, visits, pounds and individuals correlate 0.79–0.96 with each other. So they tell **one shared story**, and it is that story that differs from MMG's. This holds whether "Visits" means households (brief's definition) or individuals (The Food Group's published counts).

How we did it

- Summed 12 months per county-year; joined MMG and Census population on FIPS + year.
- Computed Spearman (and Pearson) separately for each year, at two levels: totals and per resident.
- Compared against plain population as a "no information about need" benchmark.

5. Part 2: Which metric tracks need best?

Answer: Pounds is consistently the strongest; **Individuals** is usually the weakest, with Visits close behind. The gaps between metrics are **small**, much smaller than the gap between totals and per-resident results.

	Visits	Pounds	Individuals
Totals, Spearman 2022 / 2023 / 2024	0.89 / 0.89 / 0.90	**0.92 / 0.91 / 0.92**	0.90 / 0.89 / 0.91
Per resident, Spearman	0.12 / 0.08 / −0.01	**0.20 / 0.09 / 0.02**	0.10 / 0.06 / 0.00
Per resident, Pearson	0.14 / 0.05 / −0.11	**0.32 / 0.19 / −0.02**	0.08 / −0.03 / −0.15

How sure are we? Four checks

- **Bootstrap (2,000 resamples):** for totals, Pounds beats the others in 83–97% of resamples, but only by ~0.01–0.03. Per resident, every interval for a difference includes zero. Visits vs. Individuals is never separated. → **"Consistent but small."**
- **Leave-one-out:** Hennepin pulls the totals Pearson up (without it, visits/individuals drop to ≈ 0.8 in 2024; pounds stays ≥ 0.92). Per resident, **Mahnomen** drives the only significant result: Pounds 2022 Pearson falls from 0.32 to 0.12 without it. Spearman barely moves.
- **Incomplete reporting:** removing the 7 incomplete county-years changes no correlation by more than 0.03, and the ranking of metrics stays the same.
- **Both correlation types agree** on the ordering in every year.

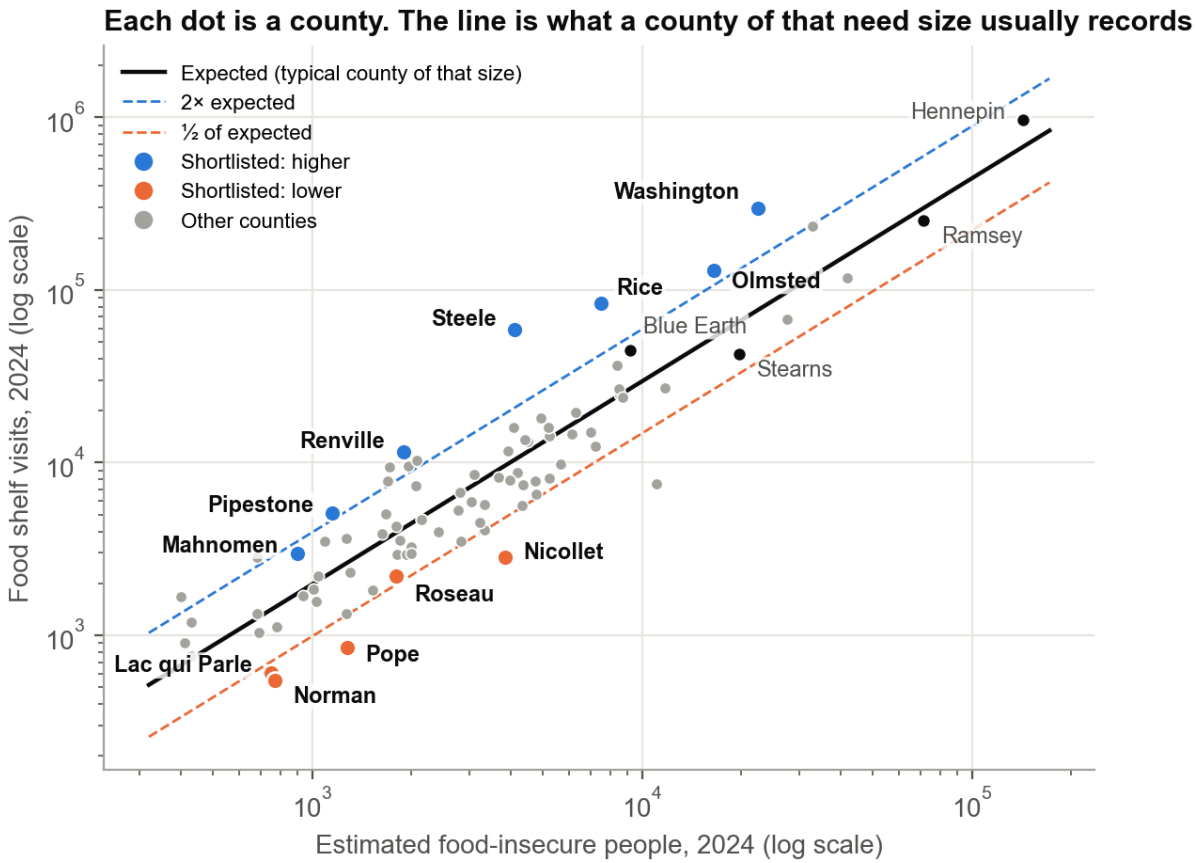
Why might pounds track need best? (talking point)

Pounds reflects how much food the food bank network moves into a county, which tends to follow population and planning, while visits and individuals depend on how often people come back and how each site counts households. That is an interpretation, not something we tested directly.

6. Part 3: Which counties break the pattern?

How "expected" is calculated (the key idea)

We can't compare raw numbers (a big county always has more). So for each year and metric, we fit a line through all counties: **activity vs. number of food-insecure people**, on a log scale. The line says what a county of that size usually records. Then each county is compared with its own point on the line.



2024 visits, 86 counties. All 12 shortlisted counties are labeled (blue = higher, orange = lower), plus four large reference counties in black. Dots above the blue dashed line are ≥2× expected; below the orange dashed line are ≤½. This shows one year and one metric; the shortlist uses all 3 years and 3 metrics, so a county (e.g. Mahnomen in 2024) can sit inside the bands here and still be shortlisted. Source: q1_county_year_mismatch.csv [§17].

Worked example (2024 visits):
Nicollet has about 3,840 food-insecure people. Counties that size typically record about **9,600** visits. Nicollet recorded **2,833** → 2,833 ÷ 9,600 = **0.30×** expected ("lower than expected").
Steele has about 4,100 food-insecure people, so expected is about **10,400**. Steele recorded **59,192** → **5.7×** expected ("higher than expected").
Same need size, roughly 20 times different activity. That's what a mismatch is.

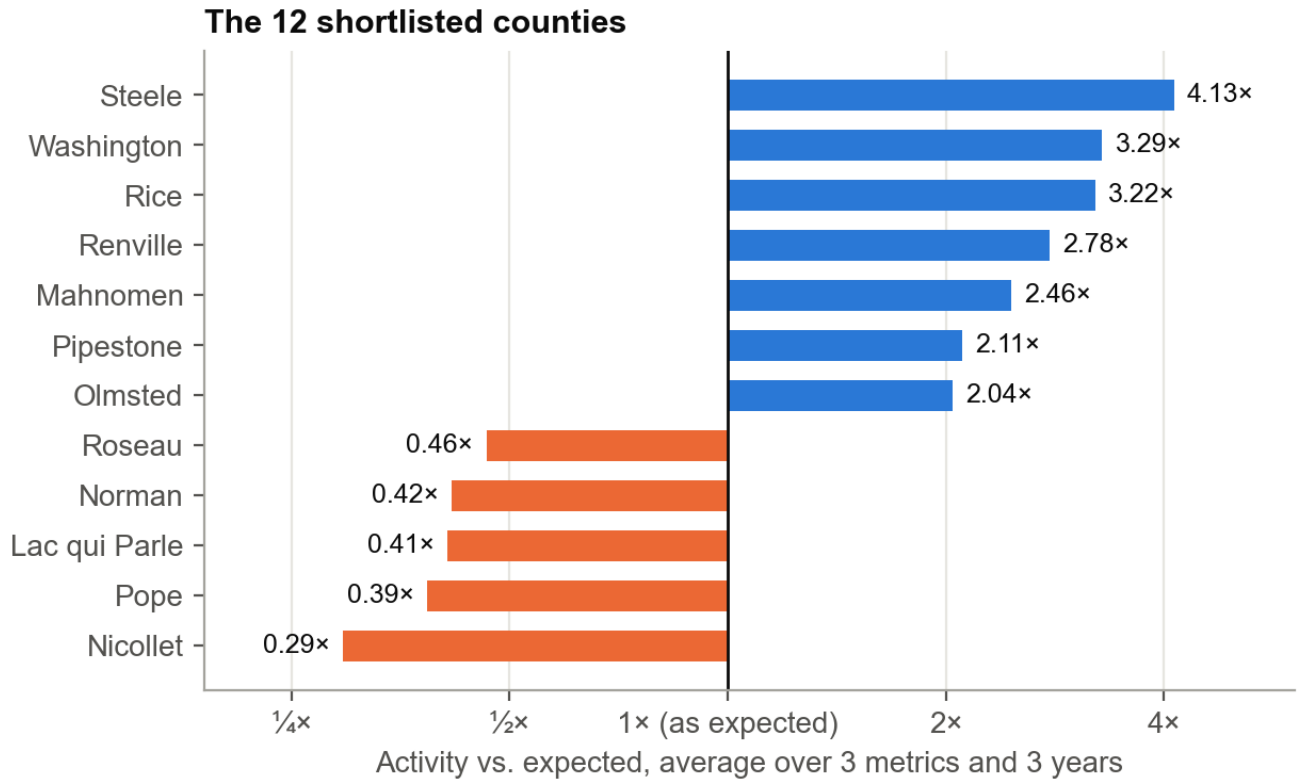
The rules

- **Flag** a county-year if activity is ≥2× or ≤½ of expected.
- **Persistent** = flagged in the same direction in ≥2 of 3 years, and never in the opposite direction.
- **Shortlist** = persistent on ≥2 of the 3 metrics, **and** confirmed by at least one alternative method, **and** an average gap of at least 2× (or ½).
- **Reporting check:** a "low" flag in a year with missing reports doesn't count toward persistence.

Why this method and not a simpler one?

We tested two alternatives: (B) per-resident activity vs. the rate, and (C) "same activity per food-insecure person everywhere". Both are biased by size (they tend to call big counties "high", size bias +0.24 to +0.34). Our method is size-neutral (−0.04 to −0.09), and all three rank counties similarly (0.86–0.92). The line's

slope is about 1.2: doubling food-insecure people goes with about 2.3× the activity.



Average of the 3 metrics × 3 years. Source: q1_county_mismatch_summary.csv [§23].

The result

- About 20% of county-years are flagged on each metric. **26 counties** are persistent on at least one metric: **10 lower, 16 higher**. **No county ever flips direction.**
- **Shortlist (12)**, all fully reported in every year:
  - Lower: Nicollet 0.29×, Pope 0.39×, Lac qui Parle 0.41×, Norman 0.42×, Roseau 0.46×. Very stable every year.
  - Higher: Steele 4.1×, Washington 3.3×, Rice 3.2×, Renville 2.8×, Mahnomen 2.5×, Pipestone 2.1×, Olmsted 2.0×. Steele and Rice are growing (Steele reached 5.7× visits, 6.3× individuals in 2024).
- **Second tier:** Benton, Cook, Kittson, Jackson (depends on method); Brown, Dodge, Mille Lacs (consistent but under 2×).
- **Possible data-quality cases, not interpreted:** Douglas, Grant, Stevens (their low flags fall in years with missing reports).
- **One-year only** (worth re-checking with 2025 data): Scott (low on all three in 2024), Wabasha (high on all three in 2024).

What "lower" and "higher" mean (and don't mean): "Lower than expected" means the county's food shelves **record** less than similar-need counties. It does **not** automatically mean people there are under-served; they may be served in a neighboring county. "Expected" is relative to other Minnesota counties, not a standard of good service.

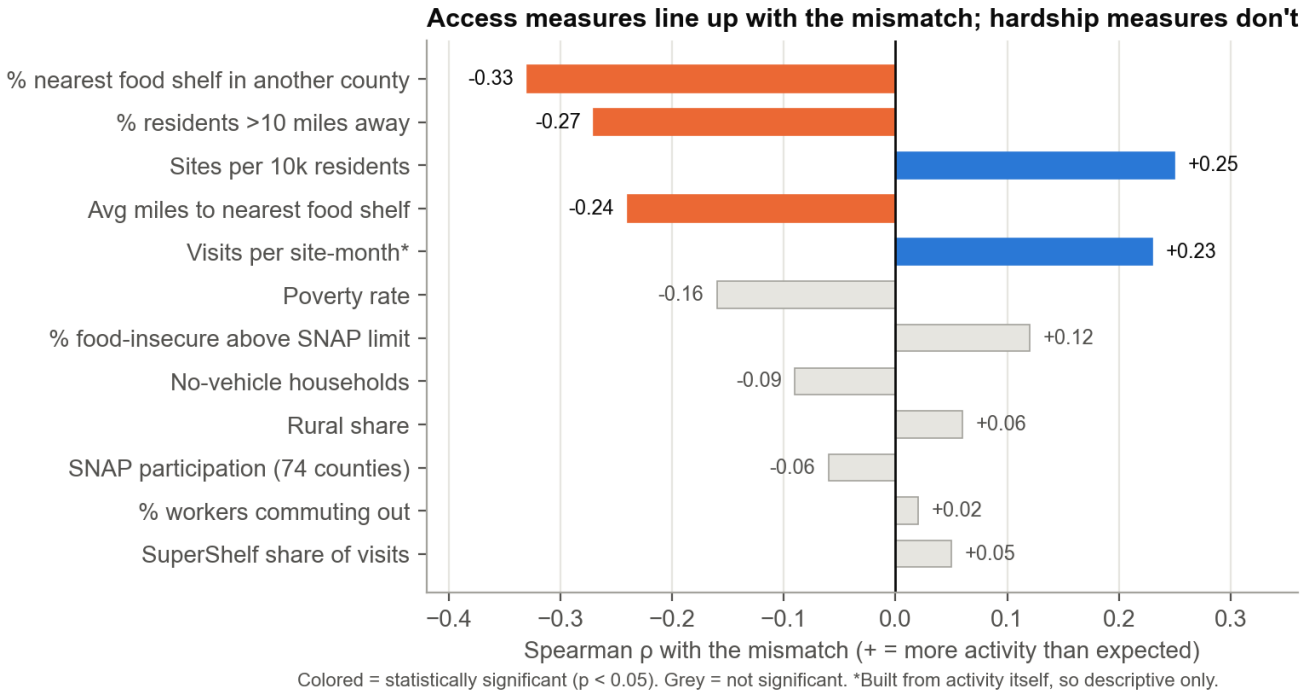
7. Part 4: What explains the mismatches?

Answer: No single factor, but **geography, access and the food bank network explain more than hardship does.** Poverty, vehicle access, rural share, SNAP, SuperShelf and commuting show little or no association. These are associations, not proof of cause.

7.1 What we tested

For each of the 86 scored counties we built a profile of public characteristics, then tested each one three ways: (1) do the lower / none / higher groups differ, (2) does it correlate with the size of the gap across all counties, (3) does it hold up in a regression with the others held constant.

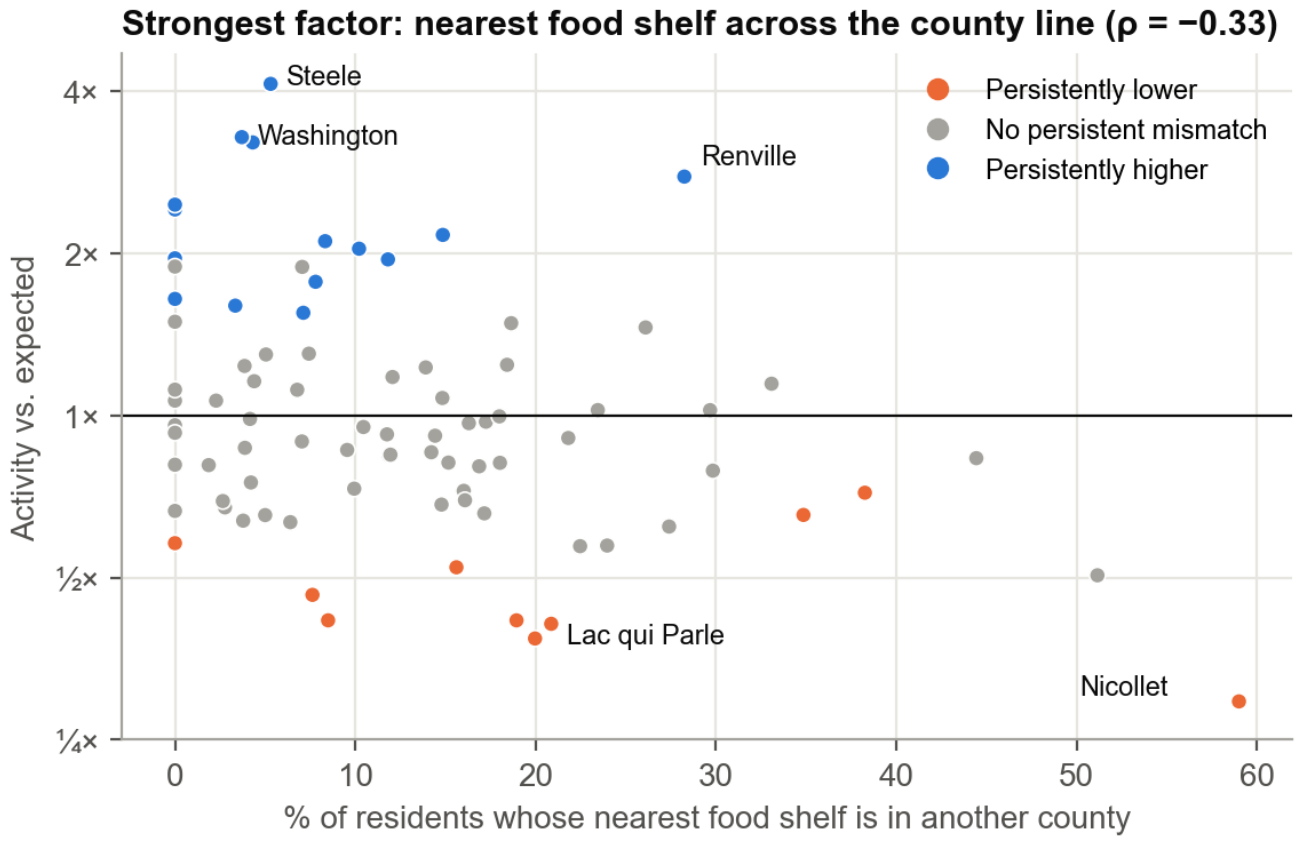
Characteristic	Source	Idea it tests
Sites per 10,000 residents	Food shelf data + ACS	Availability
Avg miles to nearest food shelf; % residents >10 miles away	Census 2020 block groups + site locations	Physical access
% of residents whose nearest food shelf is in another county	Same	Cross-county use
Poverty rate; no-vehicle households	ACS	Hardship; transport barrier
Rural share	Census 2020	Settlement pattern
% food-insecure above the SNAP income limit	MMG	People who can't get SNAP may rely more on food shelves
SNAP participation	MN SNAP (74 counties usable)	Need met by SNAP instead
Main food bank; SuperShelf share	Food shelf data; Organization.xlsx	Supply network; service model
% workers commuting to another county	Census commuting flows	Using a food shelf near work



Spearman correlation of each characteristic with every county's average gap, n = 86 (SNAP n = 74). Source: notebook [§31], [§28b].

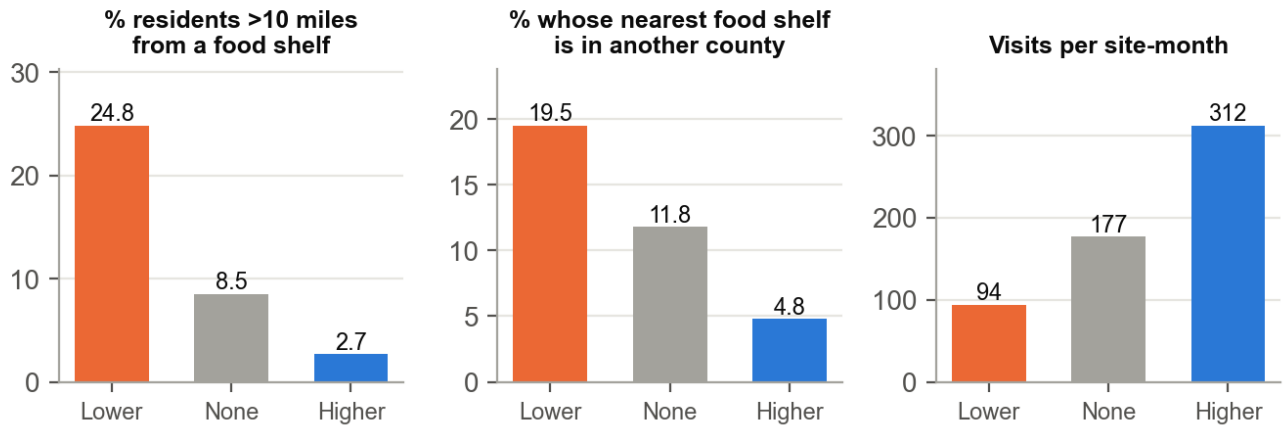
7.2 Finding: cross-county access is the strongest single factor

- The more residents whose **nearest food shelf is in another county**, the less activity the county records relative to expected: **ρ = −0.33, p = 0.002**.
- It's the **only** clear predictor in the regression (≈ −0.26 per standard deviation, p = 0.008), and it stays significant when the food bank is added (p = 0.006).
- Median across groups: **20%** (lower) vs. 12% (no mismatch) vs. **5%** (higher).



Each dot is a county. Source: q1_county_profiles.csv.

Group medians: lower-than-expected counties are farther away and have quieter sites



Group medians. Lower vs. none: p ≈ 0.04 (>10 mi), 0.05 (other county), 0.03 (visits per site). Higher vs. none, other county: p ≈ 0.04. Source: notebook [§30].

7.3 Finding: distance

- Lower-than-expected counties more often have residents far from a fixed food shelf: median **25%** beyond 10 miles vs. 9% elsewhere and 3% in the higher group.

- **Lac qui Parle, Norman, Roseau:** 30–38% of residents live more than 10 miles away, and over 90% of Norman's and Roseau's residents have their nearest food shelf in their own county, so for them it's **distance**, not crossing a county line.

How distance was measured

- **Where people live:** Census 2020 population-weighted centers of 4,706 block groups (neighborhoods of ~1,000–3,000 people).
- **Where food shelves are:** each site's Census block from Organization.xlsx → coordinates via the Census TIGERweb service. 435 fixed-location sites (food shelves, brick-and-mortar, tribal, college). Mobile pantries and home delivery excluded (their address isn't where food is handed out).
- For each block group: straight-line miles to the nearest site, and which county that site is in. Then population-weighted county averages.
- **Check:** 20 sites' blocks fell in a different county than reported (2.1% of visits). Main measure leaves them out; adding them back barely changes rankings (ρ ≥ 0.96).

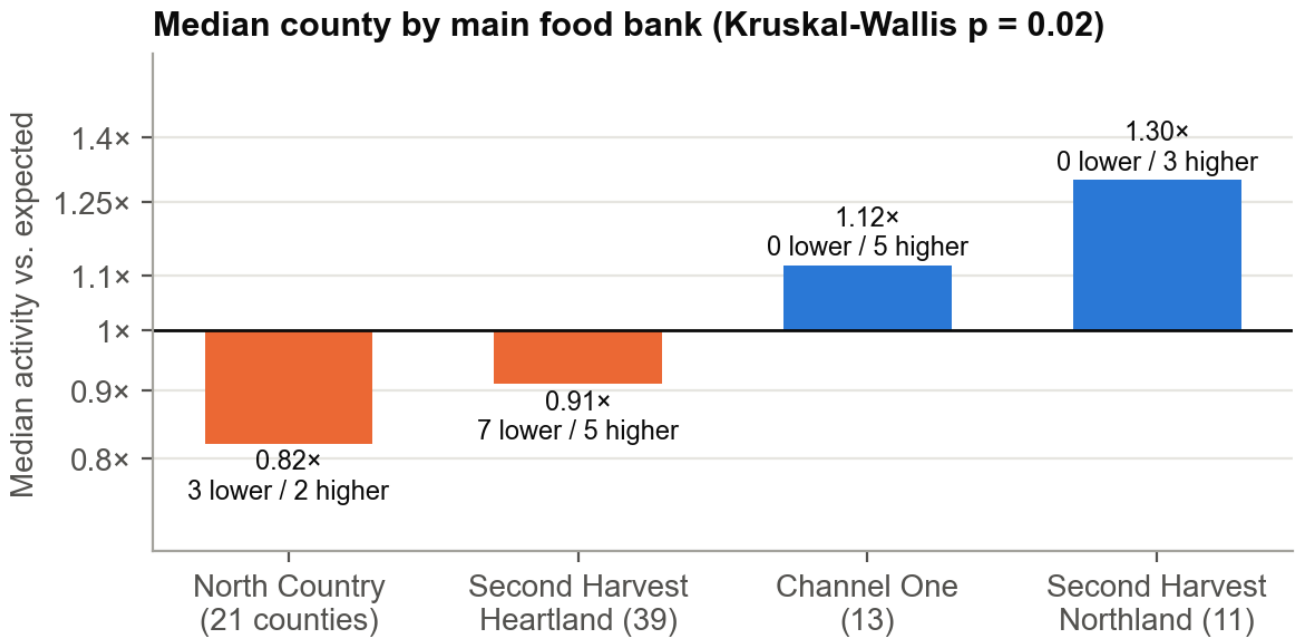
7.4 Finding: busy regional sites vs. many small sites

Activity = number of sites × activity per site. Splitting it up shows two kinds of "higher" counties:

- **Few but very busy sites, short distances:** Washington (~1,750 visits per site-month, busiest in the state), Steele (~1,160), Rice (~890), Olmsted (~920). Regional centers with low-to-median poverty.
- **Many sites for a small population:** Cook, Kittson, Jackson, Mille Lacs, Mahnomen.
- The **lower** counties mostly have **quiet** sites, not missing sites (median 94 visits per site-month vs. 177 statewide). Norman has the quietest sites in the state (~20). Only Nicollet and Benton have few sites.

Supporting pattern: many food-insecure people earn too much for SNAP [§32a]
Map the Meal Gap estimates what share of each county's food-insecure people earn **too much to qualify for SNAP**. They can't get SNAP, so a food shelf may be their main help, which would push activity above expected.
• Typical "higher" county: **36.7%** vs. **30.9%** in no-mismatch counties ($p \approx 0.15$). In the regression it points the right way (+0.18 per SD) but isn't significant ($p = 0.15$). So: **supporting, not a main finding**.
• It fits the busy hubs best: **Washington 55%, Olmsted 48%, Dodge 46%, Rice 45%** (state median 33%). **Washington has the lowest poverty on the shortlist (4.9%) yet records 3.3× expected.** Steele (35%) is near the median, so it doesn't fit; the "lower" counties are near the median too (33%), so this only helps explain the higher side.
• **How to say it:** "In the busy-hub counties, about half the food-insecure residents earn too much for SNAP, so food shelves may be their only help."

7.5 Finding: the food bank network matters (p = 0.02)

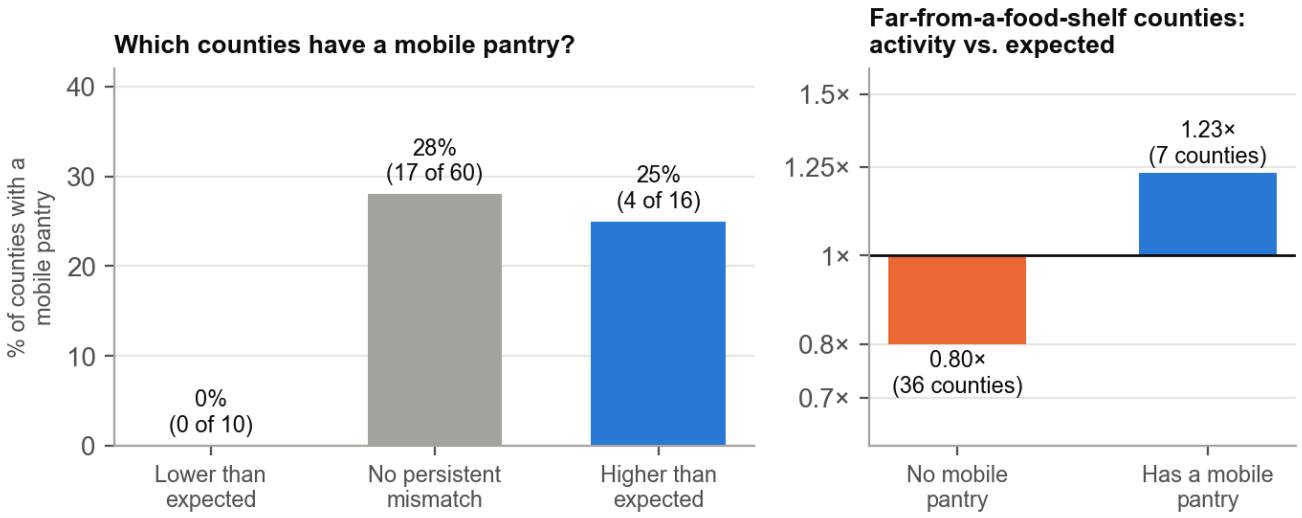


Each county assigned to the food bank supplying most of its visits. Kruskal-Wallis across food banks with 3+ counties. Source: notebook [§28b].

- Counties supplied mainly by **North Country Food Bank** (northwest MN) record less than expected (median 0.82×), including shortlisted Norman, Roseau and Pope.
- **Channel One** (southeast, Rochester) and **Second Harvest Northland** (northeast) lean higher, with **no** county below expected in either.
- Adding the food bank to the regression raises R^2 from **0.22 to 0.34**; North Country is −0.66 vs. Channel One (p = 0.02) with the other factors held constant.
- **Careful:** each food bank serves a region, so we can't separate food bank practices from other regional differences. It's a lead to discuss with The Food Group, not a conclusion. Also, Second Harvest Heartland has the most lower counties in absolute terms (7), because it serves the most counties (39).

7.6 Finding (suggestive): mobile pantries and the distance gap

A **mobile pantry** is a truck that brings food to a town on a schedule, so it can reach people who live far from a fixed food shelf. 21 of the 86 scored counties have one in the data (61 of the 93 "Outreach Programs" in Organization.xlsx are mobile).



Left: share of counties in each group with a mobile pantry (Fisher's exact test, lower vs. no mismatch: p ≈ 0.10). Right: the 43 counties with an above-median share of residents more than 10 miles from a fixed food shelf, median activity vs. expected (Mann-Whitney p ≈ 0.05). Source: notebook [§28c].

- **None of the 10 "lower" counties has a mobile pantry.** By comparison, 28% of the no-mismatch counties and 25% of the "higher" ones do. With groups this small the gap is **borderline** (Fisher's exact p ≈ 0.10), so it is suggestive, not proven.
- **Among the 43 counties where many people live far from a food shelf:** the 7 with a mobile pantry record **1.23×** expected, the 36 without one record **0.80×**. Also borderline (p ≈ 0.05), and 7 counties is a small group.
- **Cook is the clearest example:** 48% of residents live more than 10 miles away, 64% of its visits come from mobile pantries, and it records **2.4×** expected.
- **The distance-gap counties have none:** Lac qui Parle, Norman and Roseau (30–38% of residents over 10 miles away) have no mobile pantry in the data. That's a concrete, actionable option.
- **Careful:** across all counties the yes/no measure isn't significant (ρ = 0.12, p = 0.25); mobile pantry visits are part of the county's activity, so the link is partly built in; and they're recorded under the county in the reports, which may be where the program is based. Most mobile pantries are in metro counties. Present it as **support for the access story**, not a new independent factor.

7.7 Evidence that cross-county use actually happens

The food shelf data records **where the food shelf is**, not where clients live, so we can't observe cross-county use directly. But several independent pieces point the same way:

- **Nicollet → Blue Earth (strongest case).** For 59% of Nicollet residents, the nearest food shelf is in another county, mostly Blue Earth (49%, Mankato). Nicollet records 0.29× expected from very few sites (3rd percentile); Blue Earth records 1.89× from a few very busy sites (~1,110 visits per site-month). The quiet county sits right next to a busy one.
- **Wilkin (helpline call logs).** Wilkin has **zero** food shelves in the data, and MMG estimates about 720–810 food-insecure people there. Yet the helpline logged 9 calls from Wilkin, and **6 were referred to a food shelf**. Those referrals must point to food shelves in other counties. 100% of Wilkin residents' nearest food shelf is in another county (average 19 miles).
- **Lincoln and Chippewa** (low on pounds only): 35–38% of residents' nearest food shelf is in Yellow Medicine.

- **Regression:** the cross-county share predicts lower activity even with distance, rural share and poverty held constant.

Be honest about the limits: the pattern "quiet county next to a busy neighbor" holds for some pairs and not others (e.g. Benton's neighbor Stearns is itself below expected, 0.70×). Cross-county use is **plausible for some counties, not all**. Client ZIP codes would confirm it.

7.8 County-by-county explanations (what the evidence is consistent with)

County	vs. exp.	Most likely story
Nicollet	0.29×	Very few sites; 59% of residents' nearest food shelf is in another county (mostly Blue Earth, which is 1.9×). Cross-county use.
Pope	0.39×	Below-median site density, quiet sites; 20% nearest in another county; North Country network. Pounds closer to expected (0.6–0.7×).
Lac qui Parle	0.41×	Distance: 36% live >10 miles away (avg 7.0 mi); 21% nearest in Chippewa; no mobile pantry. Poverty and vehicles at/below median.
Norman	0.42×	Many sites but the quietest in the state; 30% >10 miles; no mobile pantry; North Country network. High SNAP (10.4%), weak support.
Roseau	0.46×	Distance: 38% >10 miles (avg 6.4 mi); remote; no mobile pantry; North Country network.
Steele	4.1×	Few but very busy sites (98th pct), short distances, regional center (Owatonna), rising each year. May serve people from beyond the county.
Washington	3.3×	Busiest sites in MN, everyone within 10 miles, lowest poverty on the shortlist (4.9%), 55% of food-insecure above the SNAP limit (highest).
Rice	3.2×	Busy sites (94th pct), short distances, 45% above SNAP limit. Same pattern as Steele/Washington.
Renville	2.8×	**Counterexample:** busy sites despite 45% >10 miles and 28% nearest in another county. No single factor fits.
Mahnomen	2.5×	Highest poverty (19.9%) and no-vehicle rate (8.7%), many sites, good access. MMG may understate need (hypothesis). Gap narrowed in 2024.
Pipestone	2.1×	Above-median poverty (11.6%) with busy sites.
Olmsted	2.0×	Rochester: short distances, few but busy sites (95th pct), 48% above SNAP limit.

7.9 The regression in one paragraph

One model with 7 standardized factors (site density, poverty, no-vehicle, % above SNAP limit, rural share, % >10 miles, % nearest in another county), robust standard errors, 86 counties. **$R^2 = 0.22$** (adjusted 0.15). Only the cross-county share is significant (−0.26 per SD, $p = 0.008$), roughly **0.83× expected activity for each standard deviation more residents with their nearest food shelf across the line**. Adding the main food bank: **$R^2 = 0.34$** (adjusted 0.24); the cross-county share stays significant, so geography and network each add something. **About two-thirds of the variation is still unexplained.**

Left out on purpose: average distance (nearly the same as "% >10 miles", $\rho \approx 0.9$), SNAP (would drop 12 counties), visits per site (built from activity itself).

8. What we tested and did NOT find

A "no" is still a finding. These are worth saying because judges will expect them to matter.

Factor	Result	Why it's interesting
Poverty	ρ = −0.16, p = 0.14; group medians 9.7% / 10.0% / 9.8%	MMG's model already uses economic indicators, so poverty is "inside" the expected value.
No-vehicle households	ρ = −0.09; differences within the margin of error	Mahnomen has the highest no-vehicle rate and *more* activity than expected.
Rural share	ρ = +0.06, p = 0.57	100%-rural counties are on **both** lists (Roseau, Norman, Lac qui Parle low; Cook, Kittson, Mahnomen, Renville high). **Being rural isn't the barrier; distance within rural areas is.**
SNAP participation	ρ = −0.06 (74 counties)	Combined and tribal agencies make county SNAP incomplete; no clear signal where usable.
SuperShelf certification	ρ = +0.05	Most counties have no certified sites.
Commuting to another county	ρ = +0.02	So it's about where the nearest food shelf is, not where people work.

9. Limitations (say these before the judges do)

- **Associations, not causes.** 86 counties, observational data.
- **MMG is a model estimate**, not a count, and 2024 comes from a newer release (MMG 2026) than 2022–2023 (MMG 2025).
- **No client home data.** Activity is recorded where the food shelf is. Cross-county use is inferred, not observed.
- **Straight-line distance.** Real trips are longer, especially around lakes and in the north.
- **"Individuals" counts repeat visits**, and the meaning of "Visits" is ambiguous (brief says households; The Food Group's published visits match individuals). Our conclusions hold under both.
- **SNAP is partial:** combined agencies (MNPRAIRIE, WPHS) and tribal agencies make 12 counties unusable.
- **The 2× cutoff is a judgment call.** The continuous gaps are saved so others can use a different cutoff.
- **Only 3 years** of overlap (2022–2024).
- **The food bank finding is regional:** it can't separate food bank practice from region.

10. Recommendations for The Food Group

- **Don't plan from county need estimates alone.** Use per-resident, site-level and travel-distance views.
- **Collect client ZIP codes** (even a sample) to measure cross-county use directly. This would confirm or reject the Nicollet → Blue Earth pattern.
- **Send mobile pantries to the distance gaps:** Lac qui Parle, Norman, Roseau have 30–38% of residents over 10 miles from a fixed food shelf and **no mobile pantry** in the data. Far-away counties that do have one record about 1.23× expected vs. 0.80× without (Cook: 2.4×). Next analysis: driving distance plus mobile routes.
- **Compare practices across food bank networks**, especially North Country's quiet sites (Norman, Roseau, Pope): how much food is supplied, and how visits are recorded.

- **Support busy regional hubs** (Steele, Washington, Rice, Olmsted, Blue Earth). They may be serving neighbors, and they serve many food-insecure people who earn too much for SNAP.
- **Re-check one-year cases with 2025 data:** Scott (low) and Wabasha (high).

11. Likely questions and answers

Q: Why not just compare raw totals?
Because big counties have more of everything. Population alone ranks food shelf activity as well as MMG does ($\rho \approx 0.9$ both), so totals mostly show size. That's why we also look per resident and use "expected" activity.

Q: Why Spearman instead of only Pearson?
Minnesota has one huge county (Hennepin) and many tiny ones. Pearson can be pulled by one extreme county; Spearman uses ranks so it can't. We report both; when they disagree we check which counties drive it (leave-one-out).

Q: Is "Pounds is strongest" a real result?
It's consistent (highest in every year and both methods) but small (~0.01–0.03 on totals). The bootstrap intervals overlap, so we say "consistently strongest, by a small margin", not "much better".

Q: How did you decide what's a mismatch?
We fitted activity against food-insecure people on a log scale, per year and metric. A county is flagged at 2× or ½ of the line, persistent if flagged in 2 of 3 years, and shortlisted only if ≥2 metrics agree, an alternative method agrees, and the average gap is ≥2×.

Q: Isn't the 2× cutoff arbitrary?
It's a judgment call, but a transparent one: 2× is one doubling step. We saved the continuous gaps, and the correlation analysis in Part 4 uses the continuous gaps, not the cutoff, so the explanations don't depend on it.

Q: Couldn't the mismatches just be bad data?
We checked: every shortlisted county reported all 12 months every year with at most 8% of site-months missing. No "higher" flag is driven by flagged spikes or duplicates. The three counties whose low flags coincide with missing reports (Douglas, Grant, Stevens) are labeled possible data-quality cases and not interpreted.

Q: Is cross-county use just an assumption?
It's inferred, not observed, because we don't have client addresses. But it's supported by several independent pieces: the strongest statistical factor is the cross-county share ($p = 0.002$, holds in regression); Nicollet sits next to Blue Earth which is 1.9×; and Wilkin has no food shelf yet the helpline referred 6 of its 9 callers to food shelves. We also say clearly that it doesn't fit every county (e.g. Benton/Stearns).

Q: Why doesn't poverty explain it?
MMG's food insecurity estimate already includes economic indicators like poverty and unemployment. So poverty is already "in" the expected value. What's left over is about access, not hardship.

Q: Rural counties should have less access. Why no rural effect?
Fully rural counties appear on both lists. What matters is how far people are from a food shelf, and some rural counties have many small sites close to people (Kittson, Jackson) while others don't (Roseau, Lac qui Parle).

Q: What does R^2 = 0.22 / 0.34 mean? Isn't that low?

It means these factors explain about a fifth (or a third, with the food bank) of why counties differ from expected. For county-level social data that's reasonable, and we say openly that most of the variation is unexplained, e.g. local practices, volunteer capacity, or how each site counts visits.

Q: Is the food bank finding saying North Country is doing worse?

No. Each food bank covers a region, so the difference could come from regional factors (geography, population spread) as much as from the food bank. It's a question to ask The Food Group, not a judgment.

Q: Why are rich counties like Washington so far above expected?

Washington has the lowest poverty on the shortlist (4.9%), but 55% of its food-insecure residents earn too much for SNAP, the highest share in our focus counties. Olmsted (48%), Dodge (46%) and Rice (45%) are similar. Those households can't get SNAP, so food shelves may be their main help. Higher-than-expected counties overall have more of them (37% vs. 31%, $p \approx 0.15$), so we treat it as a supporting pattern, alongside short distances and very busy sites.

Q: Do mobile pantries make a difference?

Possibly, but we call it suggestive. None of the 10 lower-than-expected counties has one (Fisher $p \approx 0.10$), and among counties where many people live far away, those with a mobile pantry record 1.23× expected vs. 0.80× without ($p \approx 0.05$, only 7 counties). Cook is the clearest case: 48% of residents over 10 miles away, 64% of visits from mobile pantries, 2.4× expected. It's borderline and partly built in (mobile visits count as activity), so we present it as support for the access story and as a recommendation, not as proof.

Q: Why is Mahnomen higher than expected?

Highest poverty (19.9%) and no-vehicle rate (8.7%) of all the focus counties, but many sites and short distances. Very high hardship plus good access. It's possible MMG understates need there (a hypothesis). It's also the county that drives the one significant per-resident correlation in Part 2.

Q: Why is Steele 4× expected?

A regional center (Owatonna) with few but very busy sites, short distances, and activity growing each year. Large sites may draw people from beyond the county. Poverty is below the state median, so it's not hardship.

Q: Why only 2022–2024?

That's the overlap between Map the Meal Gap county estimates and food shelf data. 2025–2026 have no MMG estimates yet.

Q: Why is Wilkin missing?

It has no food shelf in the data, so there's nothing to compare. We kept it in the tables, and it's actually useful evidence for cross-county use.

Q: What would you do with more time or data?

Client ZIP codes to measure cross-county use; driving distance and mobile pantry routes; compare recording practices across food bank networks; extend to 2025 when MMG releases it.

12. Suggested slide outline

#	Slide	Content / visual
1	The question	Q1 in one line; need (MMG) vs. activity (Food Group). Units differ, so we compare patterns, not subtract.
2	Same story? Partly.	Index chart (§4). Totals agree but it's size; per resident ρ ≈ 0.
3	Which metric?	Table from §5: Pounds strongest, Individuals weakest, small differences.
4	How we find mismatches	The log-log scatter with 2× / ½ lines (§6) + Nicollet vs. Steele example.
5	The 12 counties	Shortlist bar chart (§6).
6	What explains it	Factor bar chart (§7.1): access yes, hardship no.
7	Cross-county + distance	Scatter (§7.2) + Nicollet/Blue Earth + Wilkin call logs (§7.7); mobile pantry chart with Cook (§7.6).
8	Busy hubs & food bank network	Steele/Washington/Rice/Olmsted; food bank chart (§7.5) with the regional caveat.
9	What it isn't	Poverty, vehicles, rural, SNAP: no association (§8).
10	Takeaway & recommendations	One-sentence takeaway (§1) + client ZIPs, mobile pantries for distance gaps, network comparison (§10).
11	Limitations	§9, briefly.

Appendix: numbers at a glance

Item	Value
Counties scored / years	86 counties (Wilkin excluded: no food shelf) × 2022, 2023, 2024 = 258 county-years
Statewide change 2022→2024	Need +25%, visits +46%, pounds +31%, individuals +60%
Totals correlation (Spearman)	≈ 0.89–0.92; population alone ≈ 0.88–0.92
Per-resident correlation (Spearman)	−0.01 to +0.20, ≈ 0 by 2024
Metrics with each other (per resident)	ρ = 0.79–0.96
Expected-line slope / R^2	≈ 1.15–1.25 / 0.83–0.88
County-years flagged per metric	19–24%
Persistent counties (≥1 metric)	26: 10 lower, 16 higher, none switch direction
Shortlist	12: 5 lower, 7 higher
Median miles to nearest food shelf	≈ 4 miles (median county); 10% of residents' nearest is in another county
Strongest factor	% nearest food shelf in another county: ρ = −0.33 (p = 0.002); regression −0.26/SD (p = 0.008)
Food bank network	Kruskal-Wallis p = 0.02; North Country 0.82×, Heartland 0.91×, Channel One 1.12×, Northland 1.30×
Regression R^2	0.22 (adj. 0.15); with food bank 0.34 (adj. 0.24)
Food-insecure above SNAP limit (supporting)	Median 36.7% higher vs. 30.9% none (p ≈ 0.15); regression +0.18/SD (p = 0.15); Washington 55%, Olmsted 48%, Dodge 46%, Rice 45%
Mobile pantries (suggestive)	21 of 86 counties; 0 of 10 lower (Fisher p ≈ 0.10); far counties 1.23× with vs. 0.80× without (p ≈ 0.05); Cook 2.4×

Item	Value
Wilkin helpline	9 calls, 6 referred to a food shelf, 0 food shelves in the county
Data-quality cases	Douglas, Grant, Stevens

Map the Meal Gap data: Feeding America, Map the Meal Gap 2025 and 2026 releases. Census data: American Community Survey 5-year estimates, 2020 Decennial Census, ACS 2016–2020 commuting flows, TIGERweb. Food shelf, organization and helpline data: The Food Group.